Ryan McWay

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Contact Information

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University of Minnesota
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337G Ruttan Hall
Saint Paul, MN, USA 55108 – 6040

Research Interests

Primary Fields	Environmental and Resource Economics | Development Economics
Secondary Fields	Industrial Organization

Current Appointments

Ph.D. Student in Applied Economics	University of Minnesota
Principal Investigator	J-PAL MEL
Research Associate	American University of Cairo
Research Assistant	University of Minnesota
Visiting Instructor	Hamline University
Research Fellow	Institute for Replication

Education

Ph.D. in Applied Economics – University of Minnesota	Present
M.S. in Applied Economics – University of Minnesota	2026
M.S. in International and Development Economics – University of San Francisco	2020
B.S. in Economics – Saint Louis University	2018
B.S. in International Business – Saint Louis University	2018
Visiting Student	
Saint Louis University – Madrid	2017 – 2018
Bond University	2016
Spring Hill College – Bologna	2016

Publications

Job Market Paper (JMP)

Regenerative Agriculture and Farmer Profitability: Modeling Adoption for Scalable Climate Mitigation Paper Link

Abstract: Agriculture is a major source of greenhouse gas emissions, but large-scale climate mitigation in this sector depends on whether conservation practices are privately profitable for farmers. This paper studies two core regenerative practices in U.S. row-crop systems — cover cropping and no till — and asks whether adoption improves farm profitability while building soil carbon. Empirically, I combine a panel of corn, soybean, and wheat farmers in the Midwest from 2010 - 2025 with a machine-learned county-year soil panel trained on ground-truth soil tests to examine farmer decisions and outcomes to changes in the soil. I estimate reduced-form using a staggered difference-in-difference design to address endogenous adoption. I then map the estimated relationships into a structural model of farmer decision-making to evaluate adoption incentives under heterogeneous farm conditions. I find that long-run improvements in soil organic carbon (SOC) drive long-run production gains, but profitability is muted by switching costs of adoption. This suggests that large-scale climate mitigation in row crop production is limited not by agronomic potential, but by private adoption costs, which can be addressed through appropriately designed policy interventions.

Scaling Education Reform: Evidence from Morocco's Pioneer School Program Paper Link

Abstract: Can governments sustain large learning gains when evidence-based interventions move from pilot implementation to national scale? We study this question using Morocco's Pioneer School Program (PSP), a nationwide reform that combines two Global Education Evidence Advisory Panel "great buy" approaches: structured pedagogy and targeted remediation by learning level. The reform was rolled out in phases, expanding from 626 schools in 2023-24 to 4,626 schools in 2025-26, which allows us to compare effects across exposure duration and rollout cohorts. We estimate intent-to-treat effects using a matched-pair difference-in-differences design built from rich administrative records on all public primary schools and original student assessment data in Arabic, French, and mathematics. Student achievement is measured using psychometrically linked test scales (IRT), and identification is strengthened with post-double-selection LASSO controls and pre-trend evidence from seven years of historical data.

Three findings emerge. First, PSP generates large gains in foundational learning relative to comparison schools. Second, treatment effects vary across cohorts and years of exposure, consistent with changes in implementation conditions during rapid scale-up. Third, estimated impacts remain large in global comparative perspective, placing Morocco's reform among the higher-effect government-led interventions in recent evidence syntheses. Motivated by these patterns, we use the voltage-effect framework (List, 2026) to interpret how implementation quality, sample composition, and effective dosage can mediate the pilot-to-scale transition. The results contribute to the economics of education reform by showing both the promise of system-level scaling and the constraints that determine whether early gains persist as reforms expand through public delivery systems.

Journal Articles

8. Impact Evaluation of the Darsel Math Personalized Learning Platform in Jordan, joint with Caroline Krafft, Paul Glewwe, Tessa Bold, Abdulhamid Haidar, Paola Giannattasio, and Mojtaba Mahdipour **Journal of Development Economics**. Stage 1 Conditionally Accepted. Link
7. Comparing Human-Only, AI-Assisted, and AI-Led Teams on Assessing Research Reproducibility in Quantitative Social Science, joint with Abel Brodeur, David Valenta, Alexandru Marcoci, et al. **Proceedings of the National Academy of the Sciences**. 2026, 123 (22). Link
6. Reproducibility and Robustness of Economics and Political Science Research, joint with Abel Brodeur, Derek Mikola, Nikolai Cook, Edward Miguel, et al. **Nature**. 2026, 652 (8108), 151 – 156. Link
5. Spatial Polarisation. A Replication Study of Cerina et al. (The Economic Journal, 2023), joint with Noah Arman Kouchekinia and Xu Xu Cong. **Journal of Applied Econometrics**. 2026, 41 (3), 330 – 337. Link
4. The Political Consequences of Resource Scarcity: Targeted Spending in a Water-stressed Democracy. A Replication Study of Mahadevan and Shenoy (Journal of Public Economics, 2023), joint with Matthew Braaksma. **World Development Perspectives**. 2025, 39 (100707). Link
3. Market Access and Quality Upgrading: Evidence from Four Field Experiments. A Replication Study of Bold et al. (American Economic Review, 2022), joint with Karim Nchare and Pu Sun. **Journal of Comments and Replications in Economics**. 2024, 3 (2024-2). Link
2. Does Board Gender Diversity and Remuneration Impact Earnings Quality? An Empirical Analysis of Spanish Firms, joint with Paolo Saono, Laura Muro, and Pablo San Martin. **Gender in Management: An International Journal**. 2024, 39 (1), 18 – 51. Link
1. Cruising Through School: General Equilibrium Effects of Cruise Ship Arrivals on Employment and Education. **The American Economist**. 2022, 67 (1), 5 – 23. Link

Working Papers

6. Economic Contribution of Wild Pollinators: An Application of Gross-Ecosystem Product (GEP), joint with Subin Poudel and Lingling Liu. [Under Review at Ecosystem Services] Link
5. Unintended Consequences of Lockdowns, COVID-19 and the Shadow Pandemic in India. A Reproduction Study of Ravindran and Shah. (Nature Human Behavior, 2023) [Under Review at Violence Against Women] Link
4. The Statistical Significance of Statistical Insignificance. A Reproduction Study of 'The Null Result Penalty', joint with Emily Kurtz.
3. "Could You Let Me Check Your Work?" Assessing the Accessibility of Replication Materials, joint with Emily Kurtz.
2. Outsourcing Education: Experimental Evidence from Liberia. A Replication Study of Romero, Sandefur, and Sandholtz (American Economic Review, 2020), Link
1. Predicting Reproducibility and Robustness, joint with Derek Mikola and Abel Brodeur and Juan Pablo Aparicio, et al.

Works in Progress

5. Global Gross Ecosystem Product (GEP), joint with NatCap TEEMs and Chinese Academy of the Sciences' RCEES
4. What a View! The Economic Value of Scenic Quality, joint with Matthew Hockert
3. The Pollinator-Pesticide Tradeoff in Agricultural Production
2. Willingness to Pay for Biodiversity: Evidence from Citizen Science, joint with Raahil Madhok, Matthew Braaskma, and Jovin Lasway
1. Ecosystem Services from Daily Recreation and Nature-Based Tourism: An Application of Gross-Ecosystem Product (GEP), joint with Matthew Braaksma, Spencer Wood, and Lifeng Ren

Miscellaneous Research

Permanent Working Papers

3. The Impact of Early Childhood Development Interventions on Children's Health in Developing Countries: A Systematic Review and Meta-analysis of Randomized Controlled Trials, joint with Pallavi Prabhakar and Ayo Ellis. **NHH Dept. of Economics Discussion Paper**. 2022, (14/2022). Link
2. Stata Tip: Consistency and Communication in Team Dofiles, joint with James Allen IV and Hang Yu. **SSRN Working Paper Series**. 2021, (3965114). Link
1. Contrasting Incentives for Earnings Management: Board Activity and Board Remuneration in Spanish Firms, joint with Paolo Saono, Laura Muro, and Mauricio Jara-Bertin. **ResearchGate Working Paper Series**. 2019 (2.2.11103.00163). Link

Policy Briefs

3. An Evaluation of the PEPFAR-Funded Forca A Comunidade E Criancas (FCC) Program in Mozambique, joint with Dean Yang, James Allen IV, Arlete Mahumane, James Riddell IV, and Hang Yu. **USAID Feed the Future Innovation Lab for Markets, Risk, & Resilience**. 2022. Link
2. Combining Financial Incentives and Teaching to Improve COVID-19 Knowledge: A Policy Brief, joint with James Allen IV, Patricia Freitag, Faustino Lessitala, Arlete Mahumane, Laston Manja, James Riddell IV, Tanya Rosenblat, Kwasi Tabiri, Dean Yang, and Hang Yu. **University of Michigan**. 2022. Link
1. Correcting Misperceptions about Support for Social Distancing to Combat COVID-19: A Policy Brief, joint with James Allen IV, Patricia Freitag, Faustino Lessitala, Arlete Mahumane, Laston Manja, James Riddell IV, Tanya Rosenblat, Kwasi Tabiri, Dean Yang, and Hang Yu. **University of Michigan**. 2022. Link

Reports

3. Combating COVID-19 in Mozambique: Round 4 Summary Report, joint with James Allen IV, Maggie Barnard, Zach Brown, Patricia Freitag, Faustino Lessitala, Arlete Mahumane, Laston Manja, James Riddell IV, Tanya Rosenblat, Kwasi Tabiri, Dean Yang, and Hang Yu. **University of Michigan**. 2021. Link
2. Combating COVID-19 in Mozambique: Round 3 Summary Report, joint with James Allen IV, Maggie Barnard, Patricia Freitag, Faustino Lessitala, Arlete Mahumane, Laston Manja, James Riddell IV, Tanya Rosenblat, Kwasi Tabiri, Erik Tiersten-Nyman, Dean Yang, and Hang Yu. **University of Michigan**. 2021. Link
1. Combating COVID-19 in Mozambique: Round 2 Summary Report, joint with James Allen IV, Patricia Freitag, Faustino Lessitala, Arlete Mahumane, Laston Manja, James Riddell IV, Tanya Rosenblat, Kwasi Tabiri, Dean Yang, and Hang Yu. **University of Michigan**. 2020. Link

Pre-Doctoral Research Experience

J-PAL Aug. 2025 – Aug. 2027
Research Associate | American University of Cairo *Remote*
- Supervised by Dr. Paul Glewwe — Development Economics

University of Minnesota Aug. 2026 – Dec. 2026
Research Assistant | Regional Sustainable Development Partnerships *St. Paul, MN*
- Supervised by Crystal Rayamajhi — Environmental Economics

J-PAL Dec. 2024 – Dec. 2026
Principal Investigator | Morocco Innovation and Evaluation Lab *Remote*
- Supervised by Dr. Paul Glewwe — Development Economics

University of Minnesota Jan. 2026 – May 2026
Research Assistant | Department of Applied Economics *Saint Paul, MN*
- Supervised by Dr. Justin Johnson — Environmental Economics

University of Michigan Mar. 2025 – Dec. 2025
Research Associate | Ross School of Business *Remote*
- Supervised by Dr. Sarah Miller and Dr. Andrew Goodman-Bacon — Labor Economics

University of Ottawa Feb. 2024 – June 2025
Replication Fellow | Institute for Replication *Remote*
- Supervised by Dr. Abel Brodeur — Replication

Minnesota Pollution Control Agency Jan. 2025 – May 2025
Research Assistant *Saint Paul, MN*
- Supervised by David Bael and Dr. Justin Johnson — Climate Economics

University of Minnesota Aug. 2023 – Dec. 2023
Research Assistant | Department of Applied Economics *Saint Paul, MN*
- Supervised by Dr. Chengyan Yue — Neuroeconomics

University of Michigan Jun. 2020 – Aug. 2022
Research Area Specialist Associate | Population Studies Center *Ann Arbor, MI*
- Supervised by Dr. Dean Yang and Dr. Tanya Rosenblat — Development & Behavior Economics

University of San Francisco Feb. 2019 – Aug. 2020
Research Assistant | Economics Department *San Francisco, CA*
- Supervised by Dr. Jesse Anttila-Hughes — Environmental Economics

BRAC Uganda May 2019 – Aug. 2019
Research Intern | Independent Evaluation and Research Center *Kampala, Uganda*
- Supervised by Dr. Munshi Sulaiman and Dr. Danish Us Salam — Developement Economics

University of San Francisco Jul. 2018 – Jan. 2019
Business Intelligence Analyst Intern | Office of Research and Analytics *San Francisco, CA*
- Supervised by Calypso Fugit — Demography

Saint Louis University – Madrid Jan. 2018 – May 2018
Research Assistant | Business and Economics Department *Madrid, Spain*
- Supervised by Dr. Paolo Saono and Dr. Laura Muro — Business & Gender Economics

Saint Louis University Jan. 2017 – May 2017
Research Assistant | Economics Department *Saint Louis, MO*
- Supervised by Dr. Hailong Qian — Urban Economics

Teaching Experience

Instructor

QMBE 3740	Data Mining	Bachelors	Hamline University	2026
ECON 294-05	Econ. of Sustainable Development	Bachelors	Macalester College	2026
ECON 3750	Behavioral & Exp. Economics	Bachelors	Hamline University	2025
APEC 1201	App. of Excel in Econ. and Mgmt.	Bachelors	University of Minnesota	2024
————	Applied Economics Math Review	Doctoral	University of Minnesota	2023

Teaching Assistant

APEC 3611	Env. and Nat. Res. Economics	Bachelors	University of Minnesota	2026
APEC 1101	Principles of Microeconomics	Bachelors	University of Minnesota	2026, 2025
FW 8452	Conservation Biology	Doctoral	University of Minnesota	2025
APEC 5032	Economic Data Analysis	Masters	University of Minnesota	2024
APEC 3003	Intro. to Applied Econometrics	Bachelors	University of Minnesota	2023
ECON 620	Graduate Econometrics	Masters	University of San Francisco	2020
ECON 427	Applied Econometrics Capstone	Bachelors	University of San Francisco	2020
ECON 312	Intermediate Macroeconomics	Bachelors	University of San Francisco	2020
ECON 111	Principles of Microeconomics	Bachelors	University of San Francisco	2020
ECON 615	Mathematics for Economists	Masters	University of San Francisco	2019
ECON 112	Principles of Macroeconomics	Bachelors	University of San Francisco	2019

Guest Lectures

APEC 3003	Intro. to Applied Econometrics	Bachelors	University of Minnesota	2023
ECON 312	Intermediate Macroeconomics	Bachelors	University of San Francisco	2020
ECON 615	Mathematics for Economists	Masters	University of San Francisco	2019

Tutor

MATH 106	Business Statistics	Bachelors	University of San Francisco	2020, 2019
MATH 104	Algebra & Trigonometry	Bachelors	University of San Francisco	2020, 2019
ECON 372	Development Economics	Bachelors	University of San Francisco	2020, 2019
ECON 312	Intermediate Macroeconomics	Bachelors	University of San Francisco	2020, 2019
ECON 112	Principles of Macroeconomics	Bachelors	University of San Francisco	2020, 2019
BSM 302	Marketing Principles	Bachelors	University of San Francisco	2020, 2019
SOC 150	Introduction to Sociology	Bachelors	University of San Francisco	2020, 2019

Conference Presentations

CSOM Summer Econ Seminar (SumEc) Mini-Conference	2026
Camp Resources XXXII	2026
Agricultural and Applied Economics Association (AAEA) Annual Conference	2026‡
Western Economics Association International (WEAI) Annual Conference	2026
Counterfactual Methods for Policy Impact Evaluation (COMPIE) Conference	2026°
CFANS Research Symposium, University of Minnesota	2026∮, 2025∮, 2024∮, 2023∮
Sustainability Symposium, University of Minnesota	2026, 2025, 2024
Natural Capital TEEMs Annual Workshop	2025∮, 2024
Natural Capital Alliance Cook Islands Workshop	2025
Natural Capital Alliance Olympics Workshop	2025△
Mission Meets Madrid Conference – Saint Louis University Madrid	2024
Applied Social Science Associations (ASSA) Annual Meeting	2024, 2021
Upper Mississippi River Basin Ecosystem Services Project (UMRBESP) Workshop	2023∮
Nordic Conference in Development Economics (NCDE)	2023°
Annual Michigan University-Wide Sustainability & Environment (MUSE) Conference	2022, 2021
Annual International Conference on Sustainable Development (ICSD)	2021
Annual Creative Art and Research Day (CARD) Conference	2020

‡ *Session Organizer* △ *Panelist*
∮ *Poster* ◯ *Presented by Co-author*

Seminar Presentations

Department of Economics, Macalester College	2026
PhD - Economics Virtual Seminar (PhD-EVS)	2026
Morocco Innovation and Evaluation Lab (MEL), J-PAL	2026
The Natural Capital Alliance: The Earth Economy Modelers (NatCap TEEMs)	2026, 2025, 2024, 2023
Applied Economics Student Seminar Series, University of Minnesota	2026, 2023
Applied Economics Skills Workshop Series, University of Minnesota	2025⊗, 2024⊗
Service Leadership Program, Saint Louis University	2025△
Department of Economics, University of San Francisco	2024△
Community Development Branch, Federal Reserve Bank of Cleveland	2022

△ *Panelist*
⊗ *Moderator*

Seminar Organizer

PhD - Economics Virtual Seminar (PhD-EVS)	2024 – 2025
Applied Economics Skills Workshop Series, University of Minnesota	2023 – 2025
Applied Economics Student Seminar Series, University of Minnesota	2023 – 2025

Funding

Total $8,550

Internal Grants

2026	$750	UC Berkeley Haas Travel Grant *UC Berkeley Summer School in Environmental and Energy Economics*
2026	$1,000	UMN CIFAP Graduate Travel Grant *UC Berkeley Summer School in Environmental and Energy Economics*
2026	$750	NC State Camp Resources Travel Fund *Farmer Adoption of Regenerative Agriculture: A Profitable Climate Mitigation Strategy?*
2026	$1,100	UMN Applied Economics Grad Student Travel *Farmer Adoption of Regenerative Agriculture: A Profitable Climate Mitigation Strategy?*
2026	$1,500	Western Economics Association International Graduate Travel Funding *Farmer Adoption of Regenerative Agriculture: A Profitable Climate Mitigation Strategy?*
2025	$750	U. Chicago Travel Grant *Price Theory Summer Camp*
2024	$1,000	UMN CIFAP Graduate Travel Grant *NatCap Olympics*
2024	$500	UMN CFANS GSB Professional Development Opportunity Grant *American Tornadoes: Tornado Shocks on U.S. Metropolises*
2024	$500	UMN CFANS GSB Community Events & Projects Grant *Applied Economics (APEC) Student Seminar Series & Skills Workshop Series*
2024	$200	UMN COGS Community Connections Grant *Applied Economics (APEC) Masters Thesis Symposium*
2023	$500	UMN CFANS GSB Community Events & Projects Grant *Applied Economics (APEC) Student Seminar Series & Skills Workshop Series*

Recognitions

Research Awards

2025	Outstanding Paper – Emerald Literati Awards
2022	10th Michael Szenberg Prize for the Best Article Published in The American Economist
2021	Irving Fisher Graduate Research Paper Prize – Omicron Delta Epsilon
2020	Jacques Artus Award, Outstanding Thesis – University of San Francisco

Fellowships & Scholarships

2024	Graduate Scholar – Institute on the Environment (IonE), University of Minnesota
2024	Graduate Fellowship – Center for International Food and Agricultural Policy, University of Minnesota
2022	Graduate Fellowship – Applied Economics, University of Minnesota
2019	Merit Based Scholarship – University of San Francisco
2016	Boeing Bold Scholarship – Saint Louis University
	Sax International Business Scholarship – Saint Louis University
2014	Merit Based Scholarship – Saint Louis University

Miscellaneous

2018	Motley Fool College Competition Winner
2009	Eagle Scout

Professional Services and Affiliations

Editorial Services

Evaluator	The Unjournal	2024 – Present
Guest Editor	Journal of Comments and Replications in Economics	Special Issue on Payments for Ecosystem Services

Reviewer Services (# Reviews)

Journals	Environment and Social Psychology (2)
	Forum for Economics and Financial Studies (3)
	Gender in Management: An International Journal (1)
	International Review of Economics and Finance (2)
	Journal of Behavioral Economics and Policy (1)
	Journal of Business and Econometrics Studies (9)
	Journal of Infrastructure, Policy and Development (3)
	SAGE Open (2)

	Sustainable Economies (1)	
	Sustainable Social Development (1)	
Books	Bentham Science Publisher (1)	
Conferences	ATINER Annual International Conference on Tourism (1)	2024
	Association of Environmental and Resource Economists (AERE) Summer Conference (8)	2025, 2023
Grants	Council of Graduate Students (COGS), University of Minnesota (31)	2025, 2024, 2023
Affiliations		
2026	International Development Economics Association (IDEA)	
2025	Population and Environment Working Group	
2024	The Replication Network	
	Glewwe Development Lab	
	Research Fellow, Institute for Replication	
2023	American Economic Association (AEA)	
	Agricultural & Applied Economics Association (AAEA)	
	Association of Environmental and Resource Economists (AERE)	
2022	The Natural Capital Alliance: The Earth Economy Modelers (NatCap TEEMs)	
2019	Omicron Delta Epsilon (ODE)	
Service Committee Membership		
External	Digital Marketing Advisory Council – University of South Florida, College of Business	2022
	Alumni Committee – University of San Francisco, Economics Department	2022, 2021, 2020
Internal	IE Student Representative – University of Minnesota, Applied Economics	2026, 2025
	DEI Committee – University of Minnesota, ISRDI	2025, 2024, 2023
	Pre-Doc Hiring Committee – University of Michigan, Institute for Social Research	2021
	Service Leadership Advisory Board – Saint Louis University	2018, 2017, 2016, 2015

Additional Training

2026	AAEA Annual Meeting ENV Section Mentor/Peer Networking Program
	Berkeley/Sloan Summer School in Environmental and Energy Economics
2025	AMIE Mentorship Program - 5th Cohort
	Price Theory Summer Camp – University of Chicago
2024	PhD Course on Urban Economics – BREAD – IGC
	Transparency and Reproducibility Training (RT2) – UC Berkeley – BITSS
	Lecture Series Environmental Economics – IEA
2023	PhD Course on Environmental Economics – BREAD – IGC
	PhD Course on Firms and Development – BREAD – IGC
2021	World Bank Group's Manage Successful Impact Evaluation Surveys Course – World Bank
	Accessing and Analyzing Census Data Using R – SSDAN Network
2018	Minor in Italian Studies – Saint Louis University
	Minor in Service Leadership – Saint Louis University
2016	Bloomberg Marketing Concepts – Bloomberg LP

Technical Skills

Administrative:	G Suite | Git | LaTeX | Microsoft Office Suite | Prezi
Financial:	Bloomberg Marketing Concepts (BMC) | Thompson Reuters' Eikon
Languages:	Italian (Conversant) | Latin (Proficient) | Portuguese (Beginner) | Spanish (Beginner)
Statistical:	(Q)ArcGIS | Earth Engine | Julia | Python | R | Stata | SQL | Tableau
Survey Design:	ODK | SurveyCTO | Qualtrics

Professional Recommendations

Dr. Stephen Polasky
Regents Professor
Fesler-Lampert Professor of Ecological/Environmental Economics
Department of Applied Economics
University of Minnesota
Email: polasky@umn.edu

Dr. Paul Glewwe
Regents Professor
Distinguished McKnight Professor
Distinguished University Teaching Professor
Department of Applied Economics
University of Minnesota
Email: pglewwe@umn.edu

Dr. Jay Coggins
Professor
Department of Applied Economics
University of Minnesota
Email: jcoggins@umn.edu

Dr. Andrew Goodman-Bacon
Economist
Opportunity and Inclusive Growth Institute
Federal Reserve Bank of Minneapolis
Email: andrew@goodman-bacon.com